Name: C. Pavan
Regno: 192566011
Course code: CSA1603

CO4-AT-1

1. Hospital Patient Segmentation

Patient ID	Age	Monthly Medical Expense(₹)	Health Risk Score
P_1	28	12000	65
P_2	55	25000	40
P_3	42	20000	75
P_4	30	11000	70
P_5	60	28000	35

SSE values

k	SSE
1.	600
2.	350
3.	220
4.	200
5.	190

SSE → Sum of Squared

We are using elbow method

k	SSE	Reduction from Previous k
1	600	–
2	350	250
3	220	130
4	200	20
5	190	10

When $k = 2 \& 3$ The SSE value is being reduced much
when $k = 4 \& 5$ The SSE value is reducing small
This variation happen after the value $k = 3$

$\therefore$ Elbow method suggest value of $k = 3$

Based on Health risk score 3 clusters are formed

1. low risk with low Expense
2. Moderate risk with moderate Expense
3. High risk with high Expense

Impact on decision making:-

- For low risk group - Preventive care and routine monitoring
- For moderate risk
- for high risk - Intensive monitoring.

2) **E-Commerce Order Analysis:-**

Order ID	Product Category	Order Value(₹)
O_1	Electronics	55000
O_2	Clothing	25000
O_3	Electronics	57000
O_4	furniture	30000
O_5	Clothing	27000

A retailer applies Hierarchical Agglomerative Clustering (HAC) to customer order using appropriate encoded Product-Category information and standardized order value. A dendrogram is generated. Interpret how clusters are progressively formed and determine.

ie appropriate number of clusters by selecting a suitable cut-off level on the dendrogram. Explain how the selected clusters can help understand customer purchasing behaviour

HAC algorithm:-

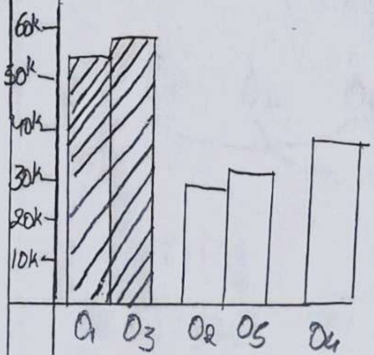
It calculate distance b/w clusters

* Merge The Two closest clusters
* Recalculate the distance
* continue it will become as small cluster

O_1 and O_3 froms the cluster 1 - Electronic

O_2 and O_5 froms the cluster 2 - Clothing

O_4 forms the cluster 3 - furniture



but the optimal no. of clusters can't be determine by the given data

order	Electronic	clothing	furniture
O_1	1	0	0
O_2	0	1	0
O_3	1	0	0
O_4	0	0	1
O_5	0	1	0

When ..

Step-2:-

Normalize order value

Use min-max normalization

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

$$O_1 = \frac{55000-25000}{57000-25000} = \frac{30000}{32000} = 0.9375$$

$$O_2 = \frac{25000-25000}{57000-25000} = 0$$

$$O_3 = \frac{57000-25000}{57000-25000} = 1$$

$$O_4 = \frac{30000-25000}{57000-25000} = 0.15625$$

$$O_5 = \frac{27000-25000}{57000-25000} = 0.0625$$

Step-3:- Calculate Ecculideon distance

O_1 and O_3 - Electronic

O_2 and O_5 - Clothing

$$d(P,Q) = \sqrt{\sum_i (P_i - Q_i)^2}.$$

$$d(O_1, O_3) = \sqrt{(1-1)^2 + (0-0)^2 + (0-0)^2 + (0.9375-1)^2}$$

$$= \sqrt{(0.0625)^2} \Rightarrow 0.0625$$

$$d(O_2, O_5) = |0 - 0.0625|$$

$$= 0.0625$$

The element within this cluster having small distance so it wir Seame cluster.

Using the some notation generate the distance matrix

	O_1	O_2	O_3	O_4	O_5
O_1	0	1.697	0.063	1.616	1.663
O_2	1.697	0	1.732	1.423	0.063
O_3	0.063	1.732	0	1.547	1.697
O_4	1.616	1.423	1.647	0	1.417
O_5	1.663	0.063	1.697	1.417	0

Step-5 :- Hierarchical Aggomerative clustering

$$D(A,B) = \min(d(x,y))$$

By seeing the smallest distance

$O_1 \quad O_2 \quad O_3 \quad O_4 \quad O_5$

$C_1 = O_1 + O_3$ $\qquad C_1 = \{O_1, O_3\} = 0.0625$

$C_2 = O_2 + O_5$ $\qquad C_2 = \{O_2, O_5\} = 0.0625$

Merging of cluster can be done by finding the distance b/w C_1 & O_4 , C_2 & O_4

3] Mobile App User Behavior - Soft Clustering

The application company wants to segment users based on:

- Daily usage time
- Login frequency
- In-app Purchase amount

The given users have different activity and Purchasing Patterns.

User	Call Duration (min)	Data usage (GB)	Recharge (₹)
U1	820	18	2000
U2	600	5	5000
U3	850	22	1820
U4	50	4	6000
U5	1320	20	1800

Probability - based membership

In soft clustering, a user does not have to belong completly to only one cluster.

for example, a user may have:

- 70% membership in cluster A
- 25% membership in cluster B
- 5% membership in cluster C

This means the user is mainly associated with cluster A but also has similarities with other groups sed on the dataset:-

- U1, U3 and U5 show high usage and frequent logins
- U2 and U4 shows lower usage and fewer logins but higher purchase values

Therefore, users can have membership probabilities based on similarities in these characteristics

Advantages of soft clustering

1. Represents real customer behavior more accurately
2. Allows a user to belong partly to multiple groups
3. Handles users with mixed behavior
4. Helps identify potential changes in user behavior
5. Supports personalized marketing

Soft clustering is useful for mobile applications because users may show characteristics of multiple behavior groups

4] Agricultural crop monitoring - Normalization

The agricultural dataset contains:

- Rainfall in mm
- Soil fertility score
- Crop yield in tons

The values have different scales:

farm	Rainfall	Soil fertility	yield
f_1	300	8	85
f_2	500	4	70
f_3	400	6	80
f_4	280	9	90
f_5	520	3	65

Before normalization

Rainfall values range from 280 to 520, while soil fertility ranges only from 3 to 9

Because rainfall has a much larger numerical scale, it can have a greater influence on distance-based clustering algorithms

After normalization

Normalization converts the features to comparable scales

for example, using min-max normalization

After normalization, the features are brought to a common scale, generally b/w 0 and 1

Normalization improves agricultural clustering because rainfall, soil fertility, and crop yield are measured on different scales. Applying normalization produces more balanced and reliable clusters for agricultural analysis mployee Performance Evaluation - Silhouette score

The company uses clustering to group employees based on:

- Productivity score
- Attendance percentage
- Skill assessment score

The given silhouette scores are

model	silhouette score
k=2	0.48
k=3	0.68
k=4	0.52

Analysis

The silhouette score measures how well data points fit within their assigned clusters

A higher silhouette score indicates

- Better separation between clusters
- Better similarity within each cluster.

Comparing the values:

- $k=2 \rightarrow 0.48$
- $k=3 \rightarrow 0.68$
- $k=4 \Rightarrow 0.52$

The highest score is 0.68 for $k=3$

Benefits of selecting $k=3$

- Provides clear employee segmentation.
- Helps identify high-performing employees.
- Helps identify employees who may need training
- Supports performance improvement programs.
- Helps managements make data-driven decisions
- makes employee evaluation more systematic

The $k=3$ model is the best choice because it has the highest Silhouette score of 0.68.